

CS 3006

Parallel and Distributed Computing

Lecture 1

Danyal Farhat

FAST School of Computing

NUCES Lahore

PDC Course Overview

Lecture's Agenda

- **Few Basic Questions**
- **Instructor's Background**
- **Classroom Etiquettes**
- **Tentative Grading Policy**
- **Administrative Information**
- **Course Learning Outcomes**

Lecture's Agenda (Cont.)

- **Textbook and Reference Books**
- **Aim of the Course**
- **Tentative Course Outline**
- **Summary**

Few Basic Questions

- **Name?**
- **CGPA (Optional)?**
- **FYP Idea?**

- **What is parallel computing?**
- **What is distributed computing?**
- **What is parallel and distributed computing?**
- **Applications that use parallel and distributed computing?**

Few Basic Questions (Cont.)

- **What is parallel computing?**

- Multiple **processors or cores** working in parallel as part of **one integrated system**.
- Normally, on a **single geographical location** with **no network** involved.

- **What is distributed computing?**

- Multiple **autonomous computers** working together as part of **one integrated system**.
- Dispersed on **multiple geographical locations** connected with the help of **network**.

Few Basic Questions (Cont.)

- **What is parallel and distributed computing?**
 - Combination of **both** parallel and distributed computing.
- **Applications that use parallel and distributed computing?**
 - Google Search Engine, **Facebook**, Netflix and many others.

Instructor's Background

- **BSCS – FAST NUCES ISB**
- **MSCS – UCP LHR**
- **Management Trainee – PTCL**
- **AM Network Support – PTCL Data Center Operations**
- **AM Network Support – PTCL Cloud Operations**
- **Cloud Specialist – Multilynx (Jazz “Garaj” Cloud Project)**
- **CS Lecturer – FAST NUCES LHR**

Classroom Etiquettes

- **80% attendance is mandatory**
 - Cross-Section class is not allowed
 - Taking two classes in the same time slot is not allowed
 - Attendance cannot be adjusted without academics office approval
 - Proxy case will have severe consequence, you may get F grade in the course
- **Cases of plagiarism will have serious consequences**
 - You may get F grade in the course or F grade in the entire semester

Classroom Etiquettes (Cont.)

- Digital **devices** should be in bag
- Note book and **pen** should be on desk
- You should have **5 – 10 A4 pages** in your bag during lecture for class activities
- Students **participation** in the class is required
 - However, talking among each other is **not appreciated**
- If a **key term** is confusing to you, raise hand and ask the instructor

Tentative Grading Policy

- Course grades will be based on **absolute** grading
- Quizzes → 10%
 - Unannounced or announced, no retake, may be best-of quizzes
- Assignments → 5%
 - Announced, no retake, no late submission, no best-of assignments
- Course Project → 10%
- Midterm I and Midterm II → 30%
- Final Exam → 45%

Administrative Information

- **Office**

- Office 55, First Floor, F Block

- **Email**

- danyal.farhat@lhr.nu.edu.pk

- **Office Hours**

- Tuesday [11:00 am – 12:00 pm]

- Thursday [11:00 am – 12:00 pm]

- **Website**

- Classroom Code (PDC BCS 7E): [byps4edh](#)

Course Learning Outcomes (CLOs)

- Explain the **architectures and principles** of parallel and distributed computing and perform analytical performance modeling.
- Implement different **parallel and distributed programming paradigms** and algorithms using Message-Passing Interface (MPI) and OpenMP.
- Analyze the **performance, scalability, and efficiency** of parallel and distributed systems (speedup, efficiency, scalability).

Textbooks

- **Introduction to Parallel Computing, Second Edition, by Ananth Grama, Anshul Gupta, George Karypis, and Vipin Kumar**
- **Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems by Martin Kleppmann**
- **Using OpenMP: Portable Shared Memory Parallel Programming, First Edition, by Barbara Chapman, Gabriele Jost, Ruud van der Pas**

Reference Books

- **Distributed Systems: Principles and Paradigms**, A. S. Tanenbaum and M. V. Steen, Prentice Hall, 4th Edition, 2024
- **Distributed and Cloud Computing: Clusters, Grids, Clouds, and the Future Internet**, K Hwang, J Dongarra and GC. C. Fox, Elsevier, 1st Ed.
- **Distributed Systems: Concept and Design** by George Coulouris, Jean Dollimore, Tim Kindberg, and Gordon Blair, 5th edition

Aim of the Course

- To understand the **fundamental concepts** of parallel and distributed computing
- Design and analysis of **parallel algorithms**
- Analyze different problems and develop **parallel programming solutions** of those problems
- Study the **challenges of Parallel and Distributed systems** and how to cope with them

Tentative Course Outline

- **Course Overview**
- **Introduction to Parallel and Distributed Computing**
- **Amdahl's Law and Karp-Flatt Metric**
- **Flynn's Classical Taxonomy & Processor to Memory Connection Strategies**
- **Network Topologies for Parallel Architectures, and Evaluation of Static Inter-connections**

Tentative Course Outline (Cont.)

- **Principles of Parallel Algorithm Design, Task Dependency Graphs, Granularity and Concurrency, and Task Interaction Graphs**
- **Decomposition Techniques**
- **Mapping Techniques**
- **Programming Shared Address Space Platforms using POSIX Thread API**
- **Shared Memory Programming with OpenMP**
- **Synchronization Constructs in OpenMP**

Tentative Course Outline (Cont.)

- **Basic Communication Operations**
- **Introduction to Distributed Systems & Types of Distributed System Architecture**
- **Distributed Operating Systems**
- **Types of Distributed Systems – Cluster Computing, Grid Computing, and Cloud Computing**
- **Programming Distributed Machines Using Message Passing Interface (MPI)**
- **Collective Communication and Computation Operations**

Tentative Course Outline (Cont.)

- **Sorting in Parallel Era**
- **Hybrid Programming (OpenMP + MPI)**
- **Distributed File System, HDFS, MapReduce and Yarn**
- **Google File System (GFS)**
- **Fault Tolerance Techniques: Hardware Redundancy, Time Redundancy, and Information Redundancy**
- **GRPC and Protocol Buffers**
- **Logical Clocks & Vector Clocks**
- **RAFT - A Consensus Algorithm for Replicated Logs**

Summary

- **Few Basic Questions**
- **Instructor's Background**
- **Classroom Etiquettes**
 - Attendance
 - Students Participation
- **Tentative Grading Policy**
 - Absolute Grading
- **Administrative Information**

Summary (Cont.)

- **Course Learning Outcomes**

- Concepts of Parallel and Distributed Computing
- Programming Paradigms - Pthread, OpenMP, and MPI
- Analytical Modelling, Algorithm dependence, and Performance Analysis

- **Textbook and Reference Books**

- **Aim of the Course**

- Fundamental Concepts of Parallel and Distributed Computing
- Design and Analysis of Parallel and Distributed Algorithms
- Challenges of PDC and their Recommended Solutions

- **Tentative Course Outline**

Questions?